Supplementary Material: When Is Periodic Detection Enough for UAV Traffic Monitoring?

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References of the form "Lemma 1" or "Section VI-F" point to the main text; numbers prefixed by S belong to this document.

S1. Proofs and Additional Theory

Assumptions A-ind, A-phase, A-stat, A-gate, A-sparse and K1–K3 are those of Sections III-A, III-C and IV-A of the main text. Numerical checks of every statement: results/p1/theory_checks.json and tests/test_p4.py in the repository.

A. Look-Count Law (Main Text, Section III-B)

Proof of Lemma 1: Look k lands on the event iff $\varphi+kS \in (b-1,\, b-1+D]$, a half-open interval of length $D = nS+fS$. Each of its n consecutive pieces of length S contains exactly one lattice point; the remainder of length fS contains one with probability f because the lattice position modulo S is uniform. This gives (i); averaging $(1-r)^N$ over (i) gives (ii); (iii) is (ii) at $r = 1$. ∎

Lemma S1 (Monotonicity): For fixed S and $r \in (0, 1]$, $D \mapsto \mathrm{miss}_r(D, S)$ is continuous, piecewise linear and non-increasing.

Proof: Between multiples of S, n is constant and (1) is affine in f with slope $-r(1-r)^n/S \le 0$; at $D = (n+1)S$ both one-sided values equal $(1-r)^{n+1}$. ∎

Proof sketch of Corollary 1: A kept censored event has $D \ge D_{obs} \ge W + 1 > S$, so both values vanish by Lemma 1(iii); (ii) follows from $D_{obs} \le D$ and Lemma S1; (iii) the selection indicator depends on b only. ∎

For $r < 1$ the duration law is also estimated by Kaplan–Meier on all non-left-censored events, which again yields an upper bound on $\mathbb{E}[\mathrm{miss}_r(D, S)]$ by Lemma S1.

Corollary S1 (Latency version): Under A-phase and A-ind, for integer S and an event of D frames,

$$P(L \le L_{max}) = 1 - \mathrm{miss}_r\big(\min(D, L_{max}+1),\, S\big). \quad (1)$$

For $r = 1$, $P(L \le L_{max}) = 1$ whenever $S \le L_{max} + 1$ and $D \ge S$; in physical units, latency $\le L_{max}/F$ seconds for every vehicle with dwell $\ge S$ frames if $f_{det} \ge F/(L_{max}+1)$.

Proof: $L \le L_{max}$ iff a successful look falls on the first $\min(D, L_{max}+1)$ frames; apply Lemma 1 to that window. ∎

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B. Kinematic Certificate and Optimal Altitude (Main Text, Section III-C)

Proof of Theorem 1: An interval of length τ contains at least $\lfloor \tau F \rfloor$ consecutive frames, among which a period-S schedule runs at least $\lfloor \lfloor \tau F \rfloor / S \rfloor = \lfloor \tau f_{det} \rfloor$ times for any phase. With $\tau \ge \ell / v_{max}$ and (2), $N \ge n$, and $\mathrm{miss} = (1-r)^N \le \varepsilon$. ∎

Remark S1: The capture rate F enters only through $f_{det} = F/S$: the requirement is on detector runs per second (design rule: detector Hz $\ge n\, v_{max}/\ell$). Vehicles clipping a footprint corner have $\ell \to 0$ and admit no finite certificate, so the certificate is stated for a region of interest crossed with chord $\ell \ge \ell_{min}$. Under A-phase, $\mathbb{E}[\mathrm{miss}] = \mathrm{miss}_r(\tau F, S)$ exactly. For an oblique camera tilted by α, $L(h) = h[\tan(\alpha + \theta/2) - \tan(\alpha - \theta/2)]$.

Let the per-look recall depend on altitude, $r = r(h)$, and define $g(h) = -\ln(1 - r(h))$ and $\Psi(h) = h\, g(h)$. Replacing n in Theorem 1 by its relaxation $\ln(1/\varepsilon)/g(h)$, the required detector rate for full traverses is

$$\tilde{f}_{det}(h) = \frac{v_{max} \ln(1/\varepsilon)}{2 \tan(\theta/2)\, \Psi(h)}, \quad (2)$$

so minimising the detector rate is maximising Ψ.

Theorem S1 (P2, existence and first-order condition): Assume (H1) $r : (0, \infty) \to (0, 1)$ continuous and (H2) $\Psi(h) \to 0$ as $h \to 0^+$ and as $h \to \infty$. Then (i) $\tilde{f}_{det}$ attains a global minimum at some $h^* \in (0, \infty)$; (ii) if r is differentiable, every interior extremum satisfies

$$\left.\frac{d \ln g}{d \ln h}\right|_{h^*} = -1, \qquad \text{and, for small } r, \quad \left.\frac{d \ln r}{d \ln h}\right|_{h^*} \approx -1; \quad (3)$$

(iii) the integer requirement $f_{det}(h) = \lceil \ln \varepsilon / \ln(1 - r(h)) \rceil\, v_{max}/(2 \tan(\theta/2) h)$ also attains a global minimum.

Proof sketch: (i) Ψ is continuous, positive and, by H2, below $\Psi(h_0)$ outside a compact $[a, b] \ni h_0$. (ii) $\Psi' = g + h g' = 0$. (iii) f_{det} is lower semicontinuous and $f_{det} \ge \tilde{f}_{det} \to \infty$ at both ends. ∎

Domain condition (corrected). Decreasing recall with $r \to 0$ is not sufficient: H2 at $h \to \infty$ requires $h\, r(h) \to 0$, i.e. recall must decay faster than $1/h$. If $r(h) \sim C h^{-\beta}$ with $\beta < 1$, $\Psi \to \infty$ and the highest admissible altitude is optimal within the model ($\beta = 1$: no interior optimum guaranteed); if r has a floor $r_0 > 0$ (e.g. a logistic curve in pixel size), $\Psi \sim h\, g(r_0) \to \infty$ and a local optimum need not be global. *Corrected statement:* an optimal altitude exists whenever recall at high altitude vanishes faster than $1/h$ and stays below one at low altitude; it is characterised by $d \ln(-\ln(1-r))/d \ln h = -1$. Otherwise the kinematic gain of altitude dominates.

Example. For $s(h) = \kappa/h$ pixels and $r(h) = r_{max}/(1 + (s_0 h/\kappa)^\beta)$, H2 holds iff $\beta > 1$, and the small-r approximation gives $h^* \approx (\kappa/s_0)(\beta - 1)^{-1/\beta}$. With $\kappa = 1{,}500$ px·m, $s_0 = 12$ px, $r_{max} = 0.9$ and $\beta = 3$: exact $h^* = 69.6$ m versus the approximation 99.2 m.

C. Gate Results (Main Text, Section IV)

Full statement of Remark 1. With $w' = \min(w, T)$ and $\Delta(r) = \mathrm{miss}_P(r) - \mathrm{miss}_G(r)$ as in the main text, the periodic schedule places $N_P \in \{0, 1\}$ looks in the window and the gate places $\mathbb{E}N_G = w' q_{in} + (T - w') q_{out}$, several of them consecutive. (i) Δ is concave on $[0, 1]$ with $\Delta(0) = 0$. (ii) Hence there is $r^\dagger \in [0, 1]$ such that Δ is non-increasing in r on $[r^\dagger, 1]$ and non-decreasing on $[0, r^\dagger]$. (iii) $r^\dagger < 1$ iff $Ta > -\mathrm{miss}'_G(1) = \mathrm{miss}_G(1)\left[\frac{w' q_{in}}{1-q_{in}} + \frac{(T-w') q_{out}}{1-q_{out}}\right]$ ($q_{in} < 1$); this always holds when $q_{in} = 1$ and $w' \geq 2$. (iv) For $q_{out} = 0$ and $w' = T$: $r^\dagger = \min\{1, (1 - (a/q_{in})^{1/(T-1)})/q_{in}\}$.

Proof sketch: miss_P is affine in r by Lemma 1 ($n = 0$, $f = Ta$); each factor $(1 - rq)^k$ is non-negative, non-increasing and convex, so their product miss_G is convex and Δ is concave with $\Delta(0) = 0$; (iii) is the condition that the maximiser is interior, and (iv) solves $\Delta'(r) = 0$. ∎

Scope. "A worse detector always favours the gate" is false in general: below $r^\dagger$ the advantage decreases with r; for small q_{in}, $r^\dagger = 1$ and Δ decreases as soon as r drops; for $Ta \geq 1$ both misses are convex and Δ has no general shape. Averaging over events preserves (i)–(ii) as long as every event has $T_e a < 1$.

Proof of Theorem 2: By Lemma 1(iii) with stride $1/(a_G + c)$, $\mathrm{miss}_P = (1 - T(a_G + c))_+$; substitute Propositions 1 and 2. ∎

Proof sketch of Corollary 2: a_G increases in q_{in}; on $\{T(a_G + c) \leq 1\}$, Δ is concave in q_{in} ($\mathcal{G}$ concave, $\mathcal{C}$ affine). (N-G) follows from $\mathcal{G} \leq 1 - T/R$ and $\mathcal{C} \geq T(1 - 1/R)(1 - \rho) q_{out}$. ∎

Proof of Corollary 3: By Lemma 1 with $Ta < 1$, $\mathrm{miss}_P = 1 - Tar$ with $a = a_G + c$, and $a_G = \rho q_{in} + (1-\rho) q_{out}$ by Proposition 1. By Proposition 2 and the union bound, $\mathrm{miss}_G = (1 - rq_{in})^{w'}(1 - rq_{out})^{T-w'} \geq 1 - r[w' q_{in} + (T - w') q_{out}]$. Subtracting gives (5). ∎

Proof sketch of Theorem 3: $q_{out,eff}(M)$ increases in M and exceeds q_{out}^{max} iff $M \geq M_{nec}$, where (N-G) of Corollary 2 fails. (ii) and (iii) follow from Proposition 1, Corollary 4 and Lemma 2. ∎

S2. Iso-KPI Boundary (Descriptive; Main Text, Criterion (ii))

Fig. S1 shows the iso-KPI boundary predicted by Theorem 3(iii) with the frozen TRAIN channel. Under its assumptions ($r = 1$, an ideal onset gate) a cue is cheaper at $\varepsilon = 2\%$ only if its per-object false-fire probability q_o lies below the curve; at the densities of the confirmatory strata the curve is low, and the moving class with its large background term q_b leaves almost no room. Because the pre-registration fixed no interval for this boundary on the real channel, the observed boundary is reported in Section S4 as a post hoc description only.

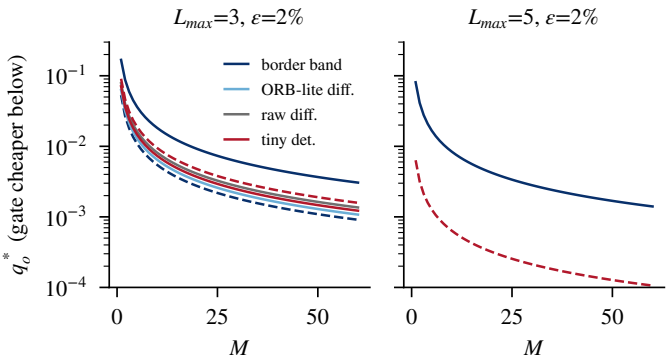


Fig. S1. Iso-KPI boundary $q_o^*(M)$ of Theorem 3(iii) (main text) for the four cues at $\varepsilon = 2\%$, computed from the frozen TRAIN channel (ρ_1, q_b per ego class, c); solid: hovering, dashed: moving. A cue is cheaper than the periodic schedule at equal latency-miss only below its curve. Theory ($r = 1$); no predicted interval was pre-registered, so this boundary is not scored.

S3. All Confirmatory Cells (Pre-Registered)

Tables S1–S3 list the 72 pre-registered cells with the frozen prediction and the UAVDT TEST result (three-way scoring of the main text, Section V-H). Source: results/p3/cells_test.csv; values are not recomputed. Table S5 repeats the cells in which the frozen model predicted a gate advantage (main text, Section VI-C).

S4. Post Hoc Analyses

All analyses in this section were designed after the test split was opened and are not part of the confirmatory family (main text, Section VI-F).

Channel drift and floors. Table S4 gives the onset-window separation at the frozen thresholds on TRAIN and TEST, Table S6 the latency-miss floors, Fig. S2 the floors and the relaxed iso-KPI comparison, and Fig. S3 the calibration of the frozen predictions.

Observed iso-KPI boundary. The observed boundary was computed with a synthetic density cue (it fires on every frame of each onset window and elsewhere with probability $1 - (1 - q_o)^{M_t}$, without refresh; the construction used to test the pipeline before freezing), on the real TEST detector dump, in three density groups formed by sequence tertiles of M. At $\varepsilon = 2\%$ the observed q_o^* is zero in every group, because the periodic floor already exceeds the target. At the relaxed target (floor plus two points) and $L_{max} = 3$ the observed values (0.0489, 0.0663, 0.0197) are larger than the closed-form points (0.0246, 0.0127, 0.0041), with intervals wide enough to contain them; at $L_{max} = 5$ and the relaxed target the observed values are 0.0254, 0.0316 and 0.0132, against closed-form points 0.0148, 0.0079 and 0.0024; the bootstrap intervals of the observed values reach zero in every group. There is no frozen interval to compare with, so this is not a criterion.

Descriptive variants. Table S7 repeats the scoring of the 72 cells under the four variants declared descriptive in advance (main text, Section VI-E).

TABLE S1
UAVDT TEST, PRE-REGISTERED ISO-KPI CELLS (H2), $\varepsilon = 2\%$: PREDICTED a_G / $a_P(\varepsilon)$ VERSUS OBSERVED; $a_P(\varepsilon) = \infty$: NO PERIODIC STRIDE REACHES ε; CI_{hi}: UPPER 95% LIMIT OF $a_G - a_P(\varepsilon)$.

cue	stratum	L_{max}	pred.	a_G pr.	a_P pr.	a_G obs.	miss_G	CI_{hi}	result
border band	hov. M 6–20	3	≤ 0	0.099	0.483	0.096	0.748	-0.580	correct
border band	hov. M 6–20	5	≤ 0	0.099	0.382	0.096	0.697	-0.559	correct
border band	hov. $M > 20$	3	≤ 0	0.099	0.457	0.096	0.893	-1000000000.000	correct
border band	hov. $M > 20$	5	≤ 0	0.099	0.316	0.096	0.849	-1000000000.000	correct
border band	mov. M 6–20	3	≤ 0	0.196	0.452	0.113	0.848	-1000000000.000	correct
border band	mov. M 6–20	5	≤ 0	0.196	0.307	0.113	0.796	-1000000000.000	correct
ORB-lite diff.	hov. M 6–20	3	≤ 0	0.201	0.483	0.205	0.647	-0.504	correct
ORB-lite diff.	hov. M 6–20	5	≤ 0	0.201	0.382	0.205	0.583	-0.446	correct
ORB-lite diff.	hov. $M > 20$	3	≤ 0	0.201	0.457	0.205	0.880	-1000000000.000	correct
ORB-lite diff.	hov. $M > 20$	5	≤ 0	0.201	0.316	0.205	0.829	-1000000000.000	correct
ORB-lite diff.	mov. M 6–20	3	≤ 0	0.292	0.452	0.272	0.705	-1000000000.000	correct
ORB-lite diff.	mov. M 6–20	5	≤ 0	0.292	0.307	0.272	0.651	-1000000000.000	correct
raw diff.	hov. M 6–20	3	≤ 0	0.197	0.483	0.182	0.538	-0.520	correct
raw diff.	hov. M 6–20	5	≤ 0	0.197	0.382	0.182	0.481	-0.458	correct
raw diff.	hov. $M > 20$	3	≤ 0	0.197	0.457	0.182	0.875	-1000000000.000	correct
raw diff.	hov. $M > 20$	5	≤ 0	0.197	0.316	0.182	0.831	-1000000000.000	correct
raw diff.	mov. M 6–20	3	≤ 0	0.377	0.452	0.387	0.605	-1000000000.000	correct
raw diff.	mov. M 6–20	5	≤ 0	0.377	0.307	0.387	0.550	-1000000000.000	correct
tiny det.	hov. M 6–20	3	≤ 0	0.233	0.483	0.213	0.540	-0.526	correct
tiny det.	hov. M 6–20	5	≤ 0	0.233	0.382	0.213	0.445	-0.518	correct
tiny det.	hov. $M > 20$	3	≤ 0	0.284	0.457	0.213	0.914	-1000000000.000	correct
tiny det.	hov. $M > 20$	5	≤ 0	0.284	0.316	0.213	0.874	-1000000000.000	correct
tiny det.	mov. M 6–20	3	≤ 0	0.217	0.452	0.345	0.751	-1000000000.000	correct
tiny det.	mov. M 6–20	5	≤ 0	0.217	0.307	0.345	0.687	-1000000000.000	correct

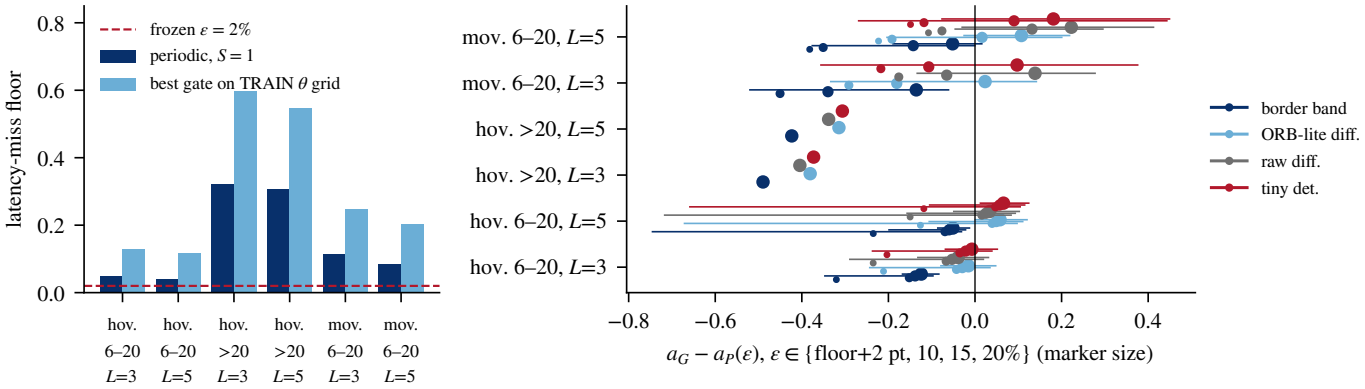


Fig. S2. Post hoc, UAVDT TEST, YOLO26-s@1024. Left: latency-miss floor of the periodic schedule run on every frame and of the best gate over the TRAIN threshold grid; dashed: the frozen target $\varepsilon = 2\%$. Right: $a_G + c - a_P(\varepsilon)$ at relaxed targets (floor plus two points, ten, fifteen and twenty percent; marker size) for the four cues at θ^*, with 95% sequence-bootstrap CI; the gate does not reach ε in any of these cells, so a negative value does not mean that the gate is a feasible cheaper option.

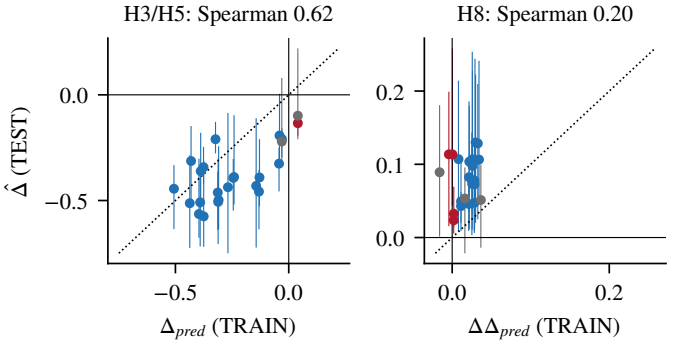


Fig. S3. Post hoc calibration of the frozen predictions on UAVDT TEST: predicted versus observed Δ (H3/H5, left) and $\Delta\Delta$ (H8, right), with 95% CI; colour: three-way result (blue correct, red wrong, grey inconclusive); dotted: identity.

TABLE S2
UAVDT TEST, PRE-REGISTERED MATCHED-COST CELLS (H3/H5): Δ_{pred} AND $\hat{\Delta}$ WITH 95% CI.

cue	stratum	L_{max}	pred.	obs.	CI	result
border band	hov. M 6–20	3	+0.040	-0.099	[-0.210, +0.220]	inconcl.
border band	hov. M 6–20	5	-0.031	-0.221	[-0.322, +0.080]	inconcl.
border band	hov. $M>20$	3	+0.040	-0.134	[-0.193, -0.098]	wrong
border band	hov. $M>20$	5	-0.030	-0.209	[-0.296, -0.158]	correct
border band	mov. M 6–20	3	-0.323	-0.210	[-0.277, -0.128]	correct
border band	mov. M 6–20	5	-0.376	-0.342	[-0.421, -0.247]	correct
ORB-lite diff.	hov. M 6–20	3	-0.241	-0.389	[-0.519, -0.097]	correct
ORB-lite diff.	hov. M 6–20	5	-0.309	-0.498	[-0.635, -0.243]	correct
ORB-lite diff.	hov. $M>20$	3	-0.245	-0.393	[-0.548, -0.303]	correct
ORB-lite diff.	hov. $M>20$	5	-0.313	-0.462	[-0.639, -0.363]	correct
ORB-lite diff.	mov. M 6–20	3	-0.437	-0.513	[-0.724, -0.286]	correct
ORB-lite diff.	mov. M 6–20	5	-0.312	-0.506	[-0.706, -0.302]	correct
raw diff.	hov. M 6–20	3	-0.041	-0.193	[-0.342, +0.005]	correct
raw diff.	hov. M 6–20	5	-0.129	-0.391	[-0.528, -0.209]	correct
raw diff.	hov. $M>20$	3	-0.043	-0.326	[-0.457, -0.250]	correct
raw diff.	hov. $M>20$	5	-0.131	-0.458	[-0.637, -0.356]	correct
raw diff.	mov. M 6–20	3	-0.268	-0.437	[-0.750, -0.086]	correct
raw diff.	mov. M 6–20	5	-0.144	-0.431	[-0.723, -0.111]	correct
tiny det.	hov. M 6–20	3	-0.431	-0.313	[-0.515, -0.147]	correct
tiny det.	hov. M 6–20	5	-0.388	-0.361	[-0.538, -0.179]	correct
tiny det.	hov. $M>20$	3	-0.507	-0.444	[-0.636, -0.333]	correct
tiny det.	hov. $M>20$	5	-0.391	-0.509	[-0.718, -0.387]	correct
tiny det.	mov. M 6–20	3	-0.375	-0.575	[-0.719, -0.311]	correct
tiny det.	mov. M 6–20	5	-0.396	-0.564	[-0.676, -0.302]	correct

TABLE S3
UAVDT TEST, PRE-REGISTERED RECALL-PAIR CELLS (H8): $\Delta\Delta_{pred}$ AND $\widehat{\Delta\Delta}$ WITH 95% CI.

cue	stratum	L_{max}	pred.	obs.	CI	result
border band	hov. M 6–20	3	+0.002	+0.024	[+0.006, +0.032]	wrong
border band	hov. M 6–20	5	+0.011	+0.043	[+0.009, +0.062]	correct
border band	hov. $M>20$	3	+0.002	+0.033	[+0.015, +0.069]	wrong
border band	hov. $M>20$	5	+0.011	+0.049	[+0.020, +0.107]	correct
border band	mov. M 6–20	3	+0.028	+0.047	[+0.006, +0.082]	correct
border band	mov. M 6–20	5	+0.028	+0.078	[+0.009, +0.125]	correct
ORB-lite diff.	hov. M 6–20	3	+0.025	+0.073	[+0.011, +0.125]	correct
ORB-lite diff.	hov. M 6–20	5	+0.025	+0.078	[+0.003, +0.143]	correct
ORB-lite diff.	hov. $M>20$	3	+0.026	+0.098	[+0.039, +0.228]	correct
ORB-lite diff.	hov. $M>20$	5	+0.026	+0.107	[+0.032, +0.254]	correct
ORB-lite diff.	mov. M 6–20	3	+0.030	+0.130	[+0.020, +0.223]	correct
ORB-lite diff.	mov. M 6–20	5	-0.004	+0.114	[+0.016, +0.199]	wrong
raw diff.	hov. M 6–20	3	+0.021	+0.046	[+0.009, +0.069]	correct
raw diff.	hov. M 6–20	5	+0.029	+0.073	[+0.011, +0.104]	correct
raw diff.	hov. $M>20$	3	+0.021	+0.082	[+0.037, +0.186]	correct
raw diff.	hov. $M>20$	5	+0.029	+0.106	[+0.036, +0.244]	correct
raw diff.	mov. M 6–20	3	+0.008	+0.107	[+0.008, +0.214]	correct
raw diff.	mov. M 6–20	5	-0.016	+0.089	[+0.001, +0.181]	inconcl.
tiny det.	hov. M 6–20	3	+0.037	+0.051	[-0.014, +0.100]	inconcl.
tiny det.	hov. M 6–20	5	+0.016	+0.053	[-0.021, +0.096]	inconcl.
tiny det.	hov. $M>20$	3	+0.034	+0.106	[+0.047, +0.241]	correct
tiny det.	hov. $M>20$	5	+0.000	+0.114	[+0.044, +0.258]	wrong
tiny det.	mov. M 6–20	3	+0.033	+0.129	[+0.025, +0.210]	correct
tiny det.	mov. M 6–20	5	+0.022	+0.104	[+0.011, +0.180]	correct

TABLE S5
THE 21 CELLS IN WHICH THE FROZEN MODEL PREDICTED A GATE ADVANTAGE ($\Delta_{pred} > \delta_{min}$), UAVDT TEST (PRE-REGISTERED). H3/H5: $\hat{\Delta} = \text{miss}_P - \text{miss}_G$ AT MATCHED COST; H8: $\widehat{\Delta\Delta} = \hat{\Delta}(\text{YOLO26N@640}) - \hat{\Delta}(\text{YOLO26S@1024})$; 95% PAIRED SEQUENCE-BOOTSTRAP CI.

hyp.	L_{max}	cue	stratum	pred.	obs.	CI	result
H3/H5	3	border band	hov. M 6–20	+0.040	-0.099	[-0.210, +0.220]	inconcl.
H3/H5	3	border band	hov. $M>20$	+0.040	-0.134	[-0.193, -0.098]	wrong
H8	3	border band	mov. M 6–20	+0.028	+0.047	[+0.006, +0.082]	correct
H8	3	ORB-lite diff.	hov. M 6–20	+0.025	+0.073	[+0.011, +0.125]	correct
H8	3	ORB-lite diff.	hov. $M>20$	+0.026	+0.098	[+0.039, +0.228]	correct
H8	3	ORB-lite diff.	mov. M 6–20	+0.030	+0.130	[+0.020, +0.223]	correct
H8	3	raw diff.	hov. M 6–20	+0.021	+0.046	[+0.009, +0.069]	correct
H8	3	raw diff.	hov. $M>20$	+0.021	+0.082	[+0.037, +0.186]	correct
H8	3	raw diff.	mov. M 6–20	+0.008	+0.107	[+0.008, +0.214]	correct
H8	3	tiny det.	hov. M 6–20	+0.037	+0.051	[-0.014, +0.100]	inconcl.
H8	3	tiny det.	hov. $M>20$	+0.034	+0.106	[+0.047, +0.241]	correct
H8	3	tiny det.	mov. M 6–20	+0.033	+0.129	[+0.025, +0.210]	correct
H8	5	border band	hov. M 6–20	+0.011	+0.043	[+0.009, +0.062]	correct
H8	5	border band	hov. $M>20$	+0.011	+0.049	[+0.020, +0.107]	correct
H8	5	border band	mov. M 6–20	+0.028	+0.078	[+0.009, +0.125]	correct
H8	5	ORB-lite diff.	hov. M 6–20	+0.025	+0.078	[+0.003, +0.143]	correct
H8	5	ORB-lite diff.	hov. $M>20$	+0.026	+0.107	[+0.032, +0.254]	correct
H8	5	raw diff.	hov. M 6–20	+0.029	+0.073	[+0.011, +0.104]	correct
H8	5	raw diff.	hov. $M>20$	+0.029	+0.106	[+0.036, +0.244]	correct
H8	5	tiny det.	hov. M 6–20	+0.016	+0.053	[-0.021, +0.096]	inconcl.
H8	5	tiny det.	mov. M 6–20	+0.022	+0.104	[+0.011, +0.180]	correct

TABLE S4
CHANNEL DRIFT AT THE FROZEN θ^*, $w = 5$ (POST HOC): ONSET-WINDOW SEPARATION $J = q_{in} - q_{out}$ ON TRAIN AND TEST, ALL SEQUENCES AND PER STRATUM (STRATUM q_{in} USES ONLY THE ONSET WINDOWS OF THE STRATUM'S EVENTS). FIRST-ORDER THRESHOLD J^* FOR $L_{max} = 3$ (SEC. VII-A OF THE MAIN TEXT).

	all		hov. M 6–20		hov. $M>20$		J^*, $L_{max}=3$	
cue	TRAIN	TEST	TRAIN	TEST	TRAIN	TEST	TRAIN	TEST
border band	+0.0362	+0.0111	+0.0103	+0.0754	+0.0777	-0.0066	0.030	0.032
ORB-lite diff.	+0.0516	+0.0135	+0.0037	+0.1057	+0.0639	-0.0287	0.114	0.119
raw diff.	+0.0576	+0.0200	-0.0136	+0.1836	+0.0475	-0.0728	0.026	0.028
tiny det.	+0.0483	+0.0077	+0.0104	+0.1744	+0.0772	-0.0493	0.147	0.154

TABLE S6
LATENCY-MISS FLOORS ON UAVDT TEST (POST HOC): PERIODIC SCHEDULE RUNNING THE DETECTOR ON EVERY FRAME ($S = 1$) AND THE BEST GATE OVER THE TRAIN THRESHOLD GRID ($R = 30$), YOLO26S@1024. A TARGET ε BELOW THE PERIODIC FLOOR CANNOT BE MET BY ANY SCHEDULE.

	$L_{max}=3$		$L_{max}=5$	
stratum (events)	periodic	gate	periodic	gate
hov. M 6–20 (121)	5.0%	12.9%	4.1%	11.6%
hov. $M>20$ (332)	32.2%	59.6%	30.7%	54.7%
mov. M 6–20 (221)	11.3%	24.7%	8.6%	20.2%

TABLE S7
DESCRIPTIVE VARIANTS OF THE 72 CONFIRMATORY CELLS ON UAVDT TEST (NOT PART OF THE CRITERIA; H2 CELLS SCORED BY SIGN ONLY).

variant	correct	wrong	inconcl.
frozen protocol (Table II)	62	5	5
earlier onset (onset$_{alt}$)	43	3	26
ROI margin 5%	63	5	4
detector score ≥ 0.05	48	1	23
detector score ≥ 0.50	63	6	3